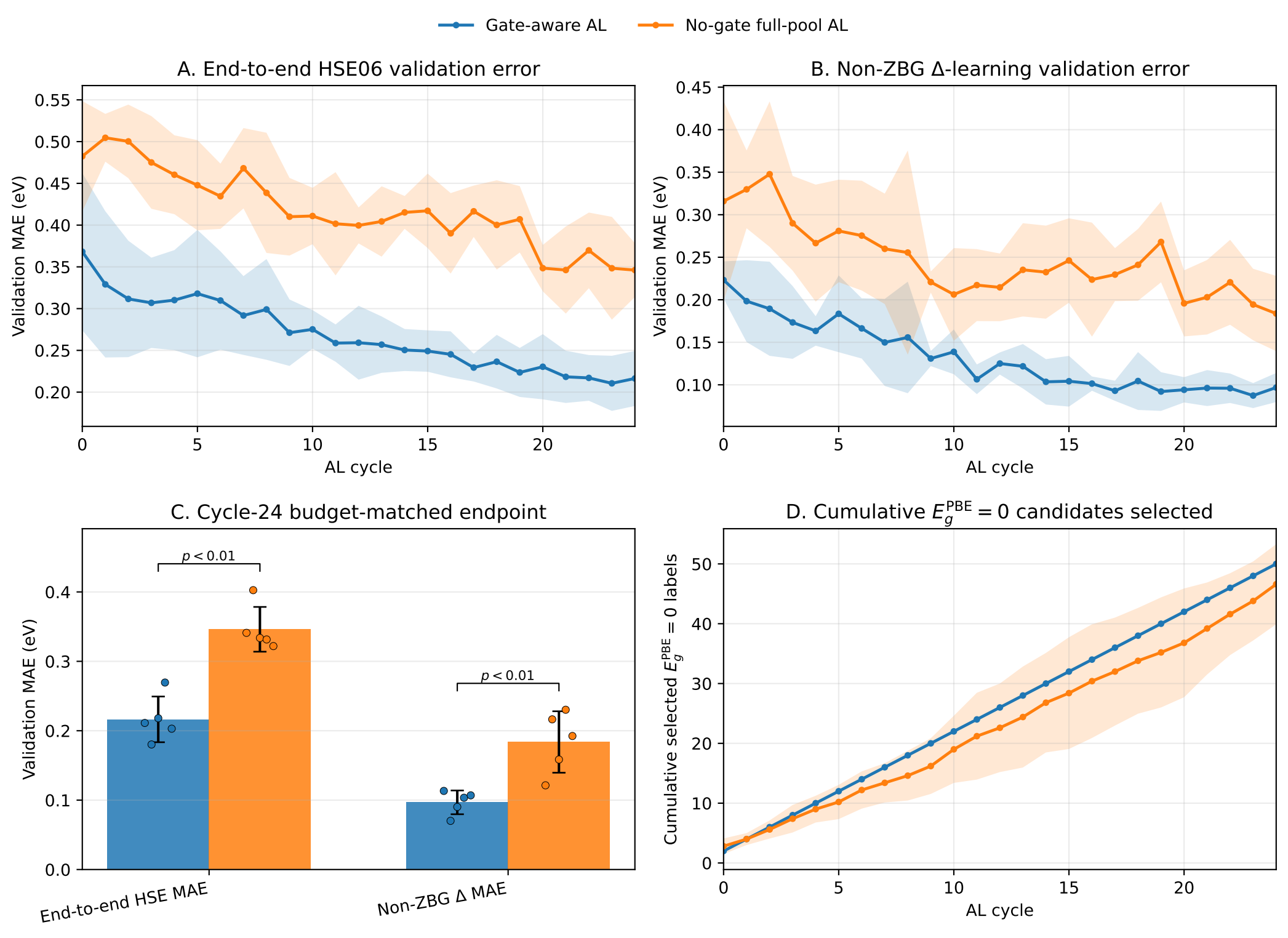
Gate-aware AL
No-gate full-pool AL
A. End-to-end HSE06 validation error
Validation MAE (eV)
AL cycle
B. Non-ZBG Δ-learning validation error
Validation MAE (eV)
AL cycle
C. Cycle-24 budget-matched endpoint
Validation MAE (eV)
p < 0.01
p < 0.01
End-to-end HSE MAE
Non-ZBG Δ MAE
D. Cumulative $E_g^{PBE} = 0$ candidates selected
Cumulative selected $E_g^{PBE} = 0$ labels
AL cycle